

## FORT ASSINIBOINE AGCM, AB, Canada

WMO: 714830

Lat: 54.410N

Lon: 114.769W

Elev: 676

StdP: 93.46

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-----|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |     |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p) |
| 12            | -31.5      | -28.3 | -35.1                         | 0.1 | -31.3 | -31.9 | 0.2 | -27.9 | 10.3                  | 3.2  | 8.7 | 1.7  | 1.2                   | 0    | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.5                   | 27.0            | 17.0 | 25.4 | 16.3 | 23.9 | 15.7 | 19.0                | 24.2 | 17.8 | 23.3 | 16.8 | 22.2 | 3.4                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.9                            | 13.1 | 22.2 | 15.6 | 12.0 | 20.6 | 14.5 | 11.2 | 19.4 | 56.8          | 23.9 | 52.8 | 23.5 | 49.6 | 22.3 | 22.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 8.6               | 7.2  | 6.0 | -36.1                      | 30.0 | 2.8                | 1.5 | -38.1                                              | 31.0 | -39.8      | 31.9 | -41.4      | 32.7 | -43.4      | 33.8 |
|                   |      |     | -36.2                      | 21.1 | 2.8                | 1.3 | -38.2                                              | 22.0 | -39.9      | 22.8 | -41.4      | 23.5 | -43.5      | 24.4 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 2.7         | -10.0 | -10.1 | -4.3  | 3.5   | 10.5  | 14.2  | 16.3  | 15.1  | 10.3  | 3.5   | -5.1  | -12.0 | (6)  |
|                                                                        | DBStd       | 11.74       | 8.81  | 8.27  | 7.50  | 4.68  | 4.36  | 3.00  | 2.57  | 2.91  | 4.28  | 4.43  | 6.57  | 7.89  | (7)  |
|                                                                        | HDD10.0     | 3265        | 620   | 563   | 443   | 199   | 47    | 4     | 0     | 2     | 47    | 205   | 453   | 683   | (8)  |
|                                                                        | HDD18.3     | 5711        | 878   | 796   | 701   | 445   | 243   | 127   | 73    | 104   | 240   | 459   | 703   | 941   | (9)  |
|                                                                        | CDD10.0     | 614         | 0     | 0     | 0     | 4     | 63    | 130   | 196   | 161   | 57    | 4     | 0     | 0     | (10) |
|                                                                        | CDD18.3     | 19          | 0     | 0     | 0     | 0     | 1     | 3     | 10    | 5     | 1     | 0     | 0     | 0     | (11) |
|                                                                        | CDH23.3     | 445         | 0     | 0     | 0     | 3     | 57    | 72    | 178   | 90    | 45    | 0     | 0     | 0     | (12) |
|                                                                        | CDH26.7     | 58          | 0     | 0     | 0     | 0     | 8     | 7     | 29    | 8     | 6     | 0     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 2.6         | 2.5   | 2.4   | 2.6   | 3.1   | 3.0   | 3.0   | 2.5   | 2.3   | 2.4   | 2.6   | 2.2   | 2.2   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 486         | 25    | 14    | 16    | 28    | 50    | 89    | 96    | 63    | 41    | 22    | 25    | 18    | (15) |
|                                                                        | PrecMax     | 705         | 98    | 35    | 37    | 60    | 91    | 154   | 212   | 102   | 112   | 60    | 76    | 43    | (16) |
|                                                                        | PrecMin     | 326         | 0     | 0     | 2     | 6     | 11    | 38    | 52    | 22    | 11    | 4     | 4     | 4     | (17) |
|                                                                        | PrecStd     | 83          | 24    | 9     | 10    | 14    | 22    | 35    | 44    | 20    | 24    | 15    | 22    | 10    | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 9.2   | 9.3   | 13.5  | 23.0  | 27.9  | 27.5  | 29.5  | 27.6  | 27.5  | 20.2  | 10.6  | 7.6   | (19) |
|                                                                        |             | MCWB        | 3.7   | 4.0   | 6.5   | 11.2  | 13.3  | 16.2  | 18.8  | 18.6  | 16.9  | 11.7  | 6.1   | 2.6   | (20) |
|                                                                        | 2%          | DB          | 6.9   | 5.5   | 10.1  | 17.8  | 24.9  | 25.4  | 27.0  | 25.7  | 24.3  | 16.2  | 7.1   | 4.2   | (21) |
|                                                                        |             | MCWB        | 3.0   | 1.6   | 3.8   | 8.1   | 12.5  | 15.1  | 18.1  | 17.9  | 15.5  | 9.1   | 3.5   | 0.8   | (22) |
|                                                                        | 5%          | DB          | 4.4   | 3.4   | 7.4   | 14.4  | 22.2  | 23.5  | 25.2  | 24.0  | 21.5  | 13.5  | 4.8   | 1.5   | (23) |
|                                                                        |             | MCWB        | 1.4   | 0.4   | 2.6   | 6.5   | 11.3  | 14.3  | 17.4  | 17.1  | 14.0  | 7.6   | 1.9   | -1.0  | (24) |
|                                                                        | 10%         | DB          | 2.0   | 1.2   | 5.0   | 11.8  | 20.0  | 21.8  | 23.5  | 22.4  | 18.6  | 11.0  | 2.7   | -1.2  | (25) |
|                                                                        |             | MCWB        | -0.4  | -1.1  | 1.3   | 5.1   | 10.3  | 13.7  | 16.5  | 16.2  | 12.3  | 6.3   | 0.8   | -3.0  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 4.1   | 4.3   | 6.8   | 11.3  | 14.5  | 18.0  | 21.1  | 20.7  | 17.7  | 12.1  | 6.8   | 3.1   | (27) |
|                                                                        |             | MCDB        | 8.2   | 8.9   | 13.0  | 22.7  | 23.7  | 24.2  | 27.3  | 25.7  | 25.5  | 19.1  | 9.8   | 7.2   | (28) |
|                                                                        | 2%          | WB          | 3.0   | 2.0   | 4.6   | 8.5   | 13.3  | 16.6  | 19.3  | 19.0  | 16.1  | 9.8   | 4.0   | 1.0   | (29) |
|                                                                        |             | MCDB        | 6.8   | 4.9   | 9.4   | 16.5  | 22.0  | 23.0  | 24.7  | 23.9  | 23.3  | 15.0  | 6.4   | 3.9   | (30) |
|                                                                        | 5%          | WB          | 1.6   | 0.6   | 2.9   | 7.0   | 12.2  | 15.4  | 18.1  | 17.7  | 14.3  | 8.0   | 2.2   | -1.0  | (31) |
|                                                                        |             | MCDB        | 4.3   | 3.2   | 7.0   | 13.4  | 20.8  | 21.6  | 23.6  | 22.6  | 20.4  | 12.7  | 4.3   | 1.2   | (32) |
|                                                                        | 10%         | WB          | -0.3  | -1.0  | 1.5   | 5.5   | 11.0  | 14.5  | 17.1  | 16.6  | 13.0  | 6.5   | 0.9   | -2.9  | (33) |
|                                                                        |             | MCDB        | 2.0   | 1.2   | 4.7   | 11.1  | 18.6  | 20.1  | 22.4  | 21.5  | 17.7  | 10.3  | 2.5   | -1.4  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 9.9   | 11.5  | 11.3  | 11.4  | 13.5  | 12.2  | 12.5  | 12.6  | 12.9  | 10.5  | 8.5   | 9.1   | (35) |
|                                                                        |             | MCDBR       | 10.7  | 12.9  | 13.6  | 17.1  | 18.3  | 15.8  | 15.8  | 15.6  | 18.2  | 15.1  | 11.0  | 11.6  | (36) |
|                                                                        | 5% WB       | MCWBR       | 7.7   | 9.5   | 8.5   | 9.2   | 8.3   | 7.3   | 8.5   | 8.7   | 10.0  | 8.9   | 7.8   | 9.1   | (37) |
|                                                                        |             | MCWBR       | 10.7  | 12.5  | 12.8  | 16.0  | 16.0  | 13.4  | 14.8  | 14.4  | 17.1  | 14.1  | 10.1  | 11.4  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.248 | 0.267 | 0.297 | 0.328 | 0.346 | 0.357 | 0.359 | 0.358 | 0.315 | 0.283 | 0.254 | 0.239 | (40) |
|                                                                        |             | taud        | 2.355 | 2.384 | 2.389 | 2.403 | 2.399 | 2.378 | 2.425 | 2.378 | 2.512 | 2.575 | 2.476 | 2.311 | (41) |
|                                                                        | Ebn at Noon |             | 737   | 844   | 889   | 902   | 900   | 890   | 881   | 859   | 862   | 824   | 735   | 663   | (42) |
|                                                                        |             | Edh at Noon | 52    | 72    | 90    | 102   | 109   | 113   | 106   | 104   | 80    | 60    | 46    | 43    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 0.85  | 1.84  | 3.27  | 4.59  | 5.52  | 5.79  | 5.83  | 4.86  | 3.37  | 1.89  | 0.89  | 0.61  | (44) |
|                                                                        | RadStd      |             | 0.07  | 0.14  | 0.27  | 0.40  | 0.45  | 0.42  | 0.38  | 0.36  | 0.35  | 0.21  | 0.09  | 0.06  | (45) |

## Historical Trends

|      |                         | Heating |         | Cooling |         |         | Degree-Days |         |         |         |         |
|------|-------------------------|---------|---------|---------|---------|---------|-------------|---------|---------|---------|---------|
|      |                         | 99% DB  | 99% DP  | 1% DB   | 1% WB   | 1% DP   | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |         |
| (46) | Station Only            | (d) N/A | (e) N/A | (f) N/A | (g) N/A | (h) N/A | (i) N/A     | (j) N/A | (k) N/A | (l) N/A | (m) N/A |
| (47) | Regional (39 neighbors) | N/A     | N/A     | N/A     | N/A     | N/A     | N/A         | N/A     | N/A     | N/A     | N/A     |

Nomenclature: See separate page